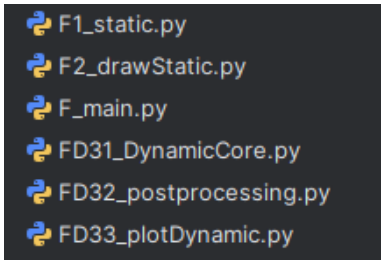
Program CleatsPy

Short user tutorial and reminder

The programme is divided into several parts (functions). The static part is calculated in the function F1_static, and the results can be viewed using F2_drawStatic. The dynamic part consists of three steps: FD31, FD32 and FD33. The first step handles dynamic settings and the core calculation, the second performs post-processing, and the last is used for plotting different results.



F_main is used when more cases are taken into consideration. For a single case, the previous function is better to use.

Geometry and materials

Initial cleats geometry is prepared in the separate <MyFileName.txt > file.

The name of the text file is inserted in the code line " FileToOpen" in the function F1_static (and later in FD31_dynamicCore). This name will be used in the file generated by the programme, where results are stored under the semi-defined name <ySR_MyFIleName.h5>.

```
if __name__ == "__main__":

    # A. Calling external file with input data about power line field
    FileToOpen = 'TestFileCleavesX.txt'
    #FileToOpen = 'primer_2.txt'
    F1_static(FileToOpen)
```

The text data in the file should be delimited by a space character. The text file includes four columns: the first column is the cleat ID, the second column is the cleat type. EP must always be at the beginning and the end, and CC must be in between. EP and CC are predefined names. The third column is the distance between cleats, and the fourth column is the division number of the span (the number of cable elements in the span). It is fine to choose an even value for easier results extraction.

The number of rows (number of cleats) is not limited, but for the sake of reasonable simulation time, keep it at 5. A sample of the txt file is presented below.

```
1 EP 0.8 2
2 CC 0.8 2
3 CC 0.8 2
4 CC 0.8 2
5 EP 0 0
```

The cable core and insulation properties are set in the subdirectory prestatic/open_menu. In the same file, the cable material and dimensions are set in the line

#vrv_c = ['CableName', vrv_material, CrossSection nominal (mm2), inner diameter (mm), outer diameter (mm)]. Refer to the figure below.

```
# predefined material properties
materials = {
    'Aluminum': {
        'E_y': 69000,  # N/mm² (69 Gpa aluminium) for solid conductor
        'rho': 2.70e-3,  # g/mm³ (2.70 g/cm³)
        'color': 'blue'
    },
    'XLPE': {
        'E_y': 43,  # N/mm² (43 MPa) for annular section
        'rho': 0.92e-3,  # g/mm³ (0.92 g/cm³)
        'color': 'green'
    },
    'Copper': {
        'E_y': 110000,  # N/mm² (110 GPa Copper) for solid conductor
        'rho': 8.96e-3,  # g/mm³ (8.96 g/cm³)
        'color': 'orange'
    }
}

# Reading cable data

#vrv_c = ['CableName',vrv_material, CrossSection nominal,inner diameter, outer diameter]
vrv_c = ['CableName', 'Aluminum', 1000, 35.7, 84]  #
```

There are two additional settings in this place. The values for contact forces, cab.kn and cab.mi are set.

```
cab = cable_class
cab.name=vrv_c[0]  # str
cab.material=vrv_c[1]  # str
cab.A=A_solid + A_pipe  # float mm2
cab.d=float(vrv_c[4])  # float mm
cab.m=mass_total  # float kg/m
cab.EI=EI_total*1e-6  # float conversion to Nm2 (from Nmm2)
cab.EA=EA_total  # float N
cab.kn=1e6#4 * Rstar**0.5 * Estar / 3 #contact stiffness (sphere)
cab.mi=0.01#0.1 # Contact damping
```

Calculate static conditions

During function F1, the static configuration is calculated and the results are written to a file (a new file the first time, overwritten on subsequent runs).

Setting dynamic conditions

Once the static conditions have been calculated, the dynamic simulation conditions are set.

In the directory dynamic, in the file d_open_menuDynamic, the settings for simulation time, short-circuit current values, subdivision of finite elements, and Rayleigh damping are defined, as shown in the figure below.

```
def d_open_menuDynamic(cab):  2 usages
    """
    Add data about time and short circuit currents
    """
    cab.tfinal = 1  # (s) whole length of simulation
    cab.tsim = 0.05  #  (s) delay of simulation from start for better plots
    cab.trelay=0.5 # (s) length of short circuit - time required to  switch off
    cab.interval = cab.tfinal # (s) time length of sub-intervals, for easier mem
    cab.ste_delitev = int(np.round(cab.tfinal / cab.interval))
    cab.note = 0.02  # (s ) interval of time for saved results

    #cab.Ik=9400  # (A) electric current Ik''
    cab.Ik = 47000  # (A) electric current Ik''
    cab.phi=89 # (degrees)surge impedance angle
    cab.Nseg=2 # number of sub-segment, minimum 2
    cab.Ncon=2 # number of contact points (checks) on element

    cab.a=30# # damping coefficient alpha
    cab.b=0.002# # damping coefficient beta
```

Running dynamic simulation

Once the dynamic simulation parameters are set, the simulation is run by FD31 (the simulation takes between 6–12 hours and depends on the size of the problem). When it is finished, to obtain the remaining data related to the system, the post-processing FD32 must be run. The data are added to the file created during the static part <ySR_MyFIleName.h5> and stored.

The results can be presented graphically using the function FD33. To do this, the file name should be written into file_names. The span number or cleat number is selected by setting the value gr.k. The number of files to be compared, gr.n_files, is set to one in this version (do not change it). The number of curves to plot, gr.n_curves, is also set to one in this version (do not change it). Choose the figure-graph type to be plotted with gr.plot_type = Y, where Y can be 1, 2, 3, 4, or 5. See the figure below.

```
# File names
file_names = ['primer_1.txt', 'TestFileCleavesX.txt', 'primer_5.txt',
              None, None]

gr=plot_class
# User parameters
gr.k = 3  # span or cleave index to plot  (first span is 1, internally to rea
gr.n_files= 1 # number of files to be opened and compared, Temporary only one
gr.n_curves = 1  # number of curves to plot, that is used if only one file is
gr.plot_type = 2  # type of graph

# plot type
# 0- xyz by component displacement, cable in the middle of span, dimensional
# 1- normalised ABSOLUTE displacement in the middle of span
# 2 Forces by components x,y,z in Cleats
# 3 Forces internal in cable by components x,y,z in Midspan
# 4 Displacements x,z along cable line y by time steps
# 5 ForcesInsideCleat- attempt not finished yet
```